

AGILE SOFTWARE ENGINEERING (CS303)

Story Point Estimation Techniques

Learning Objectives

Understand Story Points, Agile estimation, Planning Poker, T-Shirt Sizing, Fibonacci sequence, Velocity, Sprint Capacity, and estimation best practices.

Introduction

Estimation is an important Agile activity used to predict the effort required to complete user stories. Instead of estimating in hours, Agile teams often use Story Points.

What are Story Points?

Story Points are a relative unit of measurement used to estimate the effort, complexity, uncertainty, and risk associated with a user story.

Why Use Story Points?

Story Points focus on relative effort rather than exact time, making estimates more flexible and realistic.

Factors Considered in Story Point Estimation

1. Complexity of the task
2. Amount of work involved
3. Technical uncertainty
4. Risk factors
5. Team experience

Fibonacci Sequence Estimation

Many Agile teams use the Fibonacci sequence for estimation: 1, 2, 3, 5, 8, 13, 21, 34, 55. Larger values indicate greater effort and uncertainty.

Planning Poker

Planning Poker is a collaborative estimation technique where team members independently choose estimation cards and then discuss differences before reaching consensus.

Steps in Planning Poker

1. Product Owner presents the story.
2. Team discusses requirements.
3. Members select estimation cards.
4. Estimates are revealed simultaneously.
5. Differences are discussed.
6. Final estimate is agreed upon.

T-Shirt Sizing Technique

Stories are categorized using sizes such as XS, S, M, L, XL based on relative effort. This method is quick and easy for high-level estimation.

Affinity Estimation

Affinity Estimation groups similar stories together and arranges them according to complexity and effort.

Velocity

Velocity measures the number of Story Points completed by a team in a Sprint. It helps forecast future Sprint capacity.

Sprint Capacity

Sprint Capacity refers to the amount of work a team can realistically complete during a Sprint based on available resources and previous performance.

Best Practices

- Use team collaboration for estimates.
- Focus on relative effort.
- Avoid converting Story Points directly to hours.
- Re-estimate when requirements change.
- Use historical Velocity data.

Worked Example

If Story A is assigned 3 points and Story B appears twice as complex, Story B may be assigned 8 points using Fibonacci estimation.

Advantages of Story Point Estimation

Improves planning accuracy, encourages collaboration, supports Sprint planning, and helps manage uncertainty.

Exam Important Questions

1. Define Story Points.
2. Explain Planning Poker.
3. Discuss Fibonacci Estimation.
4. What is Velocity in Agile?
5. Explain T-Shirt Sizing Technique.

Summary

Story Point Estimation helps Agile teams estimate effort effectively using techniques such as Planning Poker, Fibonacci estimation, T-Shirt Sizing, and Velocity tracking.